Real-Time Video Inference on 8× NVIDIA RTX PRO 5000

MiniMax H3 with vLLM-Omni

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Real-Time 768p: The Full Pipeline Must Fit Within the Clip Duration

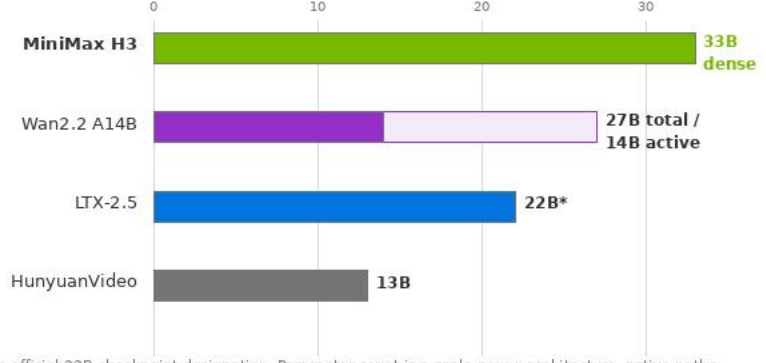
For 362 frames at 24 FPS, the encoder, eight DiT steps, communication, and VAE decode share one 15.08 s end-to-end budget.

REAL-TIME DEFINITION

RTF = end-to-end latency ÷ media duration ≤ 1.0

768p · 362 frames @ 24 FPS · 15.08 s media → ≤ 15.08 s end to end

PUBLISHED VIDEO DENOISER SCALE · BILLION PARAMETERS



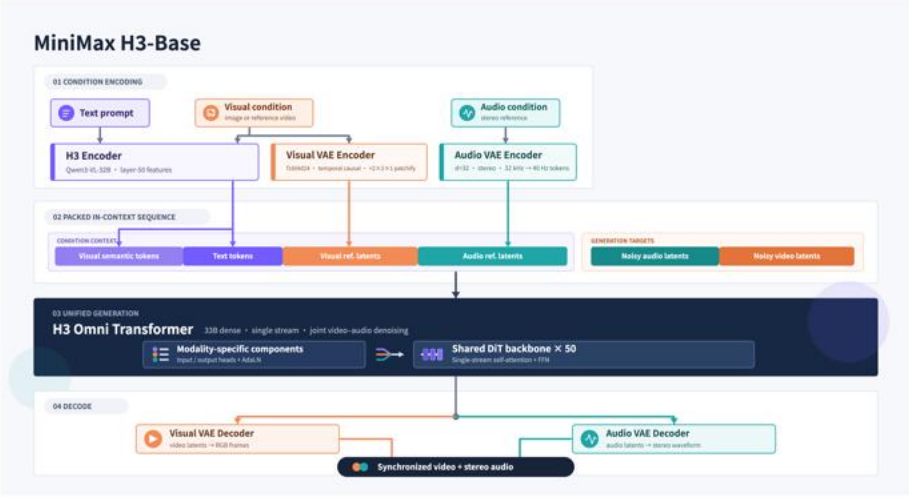
*LTX-2.5 uses the official 22B checkpoint designation. Parameter count is a scale proxy; architecture, active paths, sequence length, and denoising steps determine actual FLOPs.

Largest dense video transformer in this peer set: 33B

At native 768p, the dense generator operates over long audio-video sequences; the initial open release uses full attention, making tensor throughput and memory movement first-order serving constraints.

MINIMAX H3-BASE · OFFICIAL ARCHITECTURE

32B ENCODER → 33B DENSE DiT × 50 → VISUAL + AUDIO VAE



MODEL RESIDENCY

≈130 GB

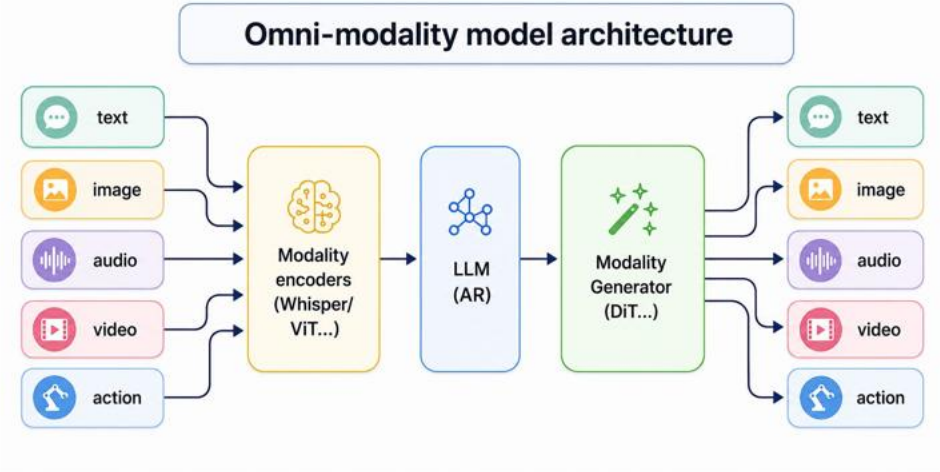
raw BF16 weights for the 32B encoder + 33B generator—about 1.8× one 72 GB GPU—before Visual/Audio VAE, activations, attention workspace, and graph pools

Stage placement, precision, offload, and workspace reuse are therefore part of performance. About 13B H3 AdaLN parameters can be precomputed and cached for inference-only deployment.

Sources: official MiniMax H3, LTX-2.5, HunyuanVideo and Wan2.2 repositories · author 362-frame workload

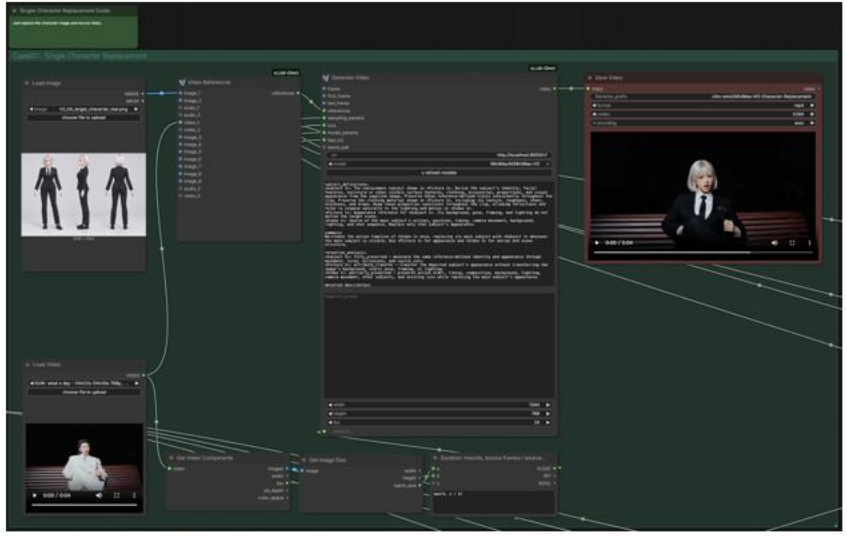
vLLM-Omni Connects Visual Workflows to Multi-GPU Serving

ComfyUI remains the authoring surface; vLLM-Omni owns serving, stage placement, batching, and multi-GPU execution



OFFICIAL vLLM-OMNI ARCHITECTURE

One serving layer spans omni-modal inputs, autoregressive reasoning, and DiT generation.



COMFYUI WORKFLOW · PR #9194

MiniMax-H3 character replacement calls the vLLM-Omni Generate Video endpoint; PR #9194 remains open.

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

Frontier Models · The Cards

One Blackwell architecture (CC 12.0/12.1) spans several very different deployment envelopes

SKU	CC / SMs	Memory	Power	Interconnect
RTX PRO 5000 Blackwell 8-GPU baseline	12.0 / 110	72 GB GDDR7 ECC 48 GB variant	**350 W configured*** **300 W public TBP**	PCIe Gen5 ×16 no GPU NVLink
RTX PRO 6000 Blackwell WS / Server	12.0 / 188	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 no GPU NVLink
GeForce RTX 5090	12.0 / 170	32 GB GDDR7	575 W	PCIe Gen5 ×16 NVLink: no
DGX Spark · GB10	12.1 / 48	128 GB unified LPDDR5X	140 W SoC 240 W system	NVLink-C2C in SoC 200 Gb/s ConnectX-7

SHARED BY DISCRETE CARDS · BLACKWELL FP4 · GDDR7 · PCIe GEN5 · NO GPU-TO-GPU NVLINK

Watch sustained clocks, not peak TOPS: controlled tensor-heavy tests reported SwPowerCap in 94.8–100% of valid samples; one rank stayed at 1.942 GHz and ran the same fine-attention input 6.78% slower. A 600 W-class design offers more headroom, but still requires a matched chassis A/B.

*350 W is the local configured/enforced limit; NVIDIA publicly specifies 300 W TBP for RTX PRO 5000.

Sources: NVIDIA product specifications · CUDA GPU Compute Capability · local system inventory and controlled clock study